



Human pose estimation

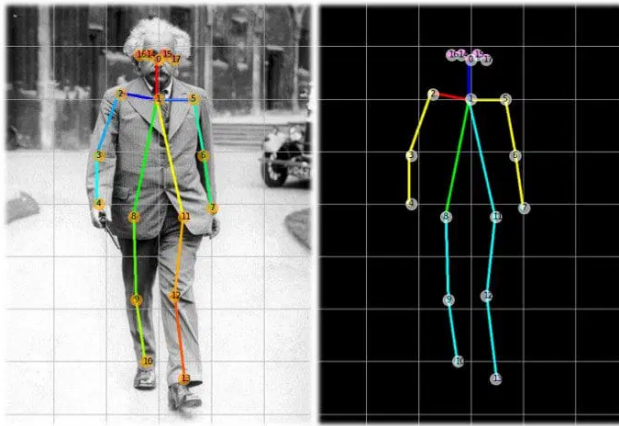
Laboratory of robotics and control systems
ITT PROJECT MANAGEMENT SERVICES L.L.C

Plan

1. What is human pose estimation
2. Datasets
3. Architectures
4. Metrics

Task definition

Identify the key points on the human body, which can help us determine the pose of a person

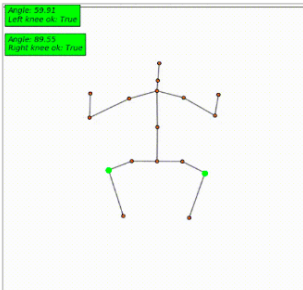


Human Pose estimation flow

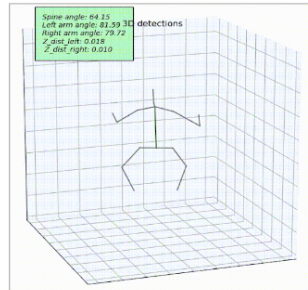
Original image



Result of 2D Pose Estimation work



Result of 3D Pose Estimation work



- Extract 2D key points
- Convert 2D key points into 3D, adding the depth dimension.

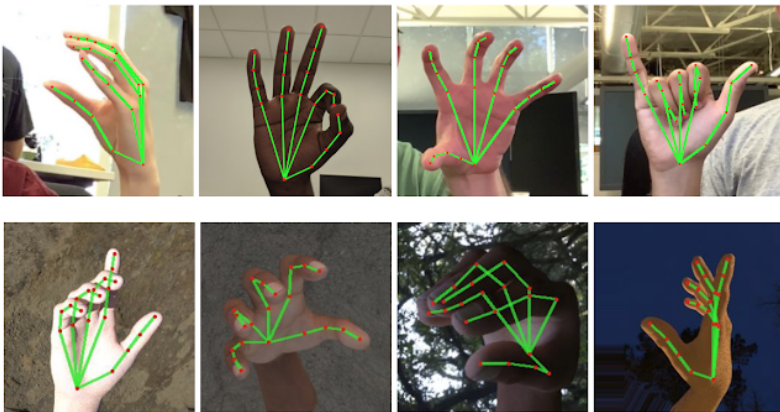
Application: fitness assistant



- Correct your technique
- Prevent injuries



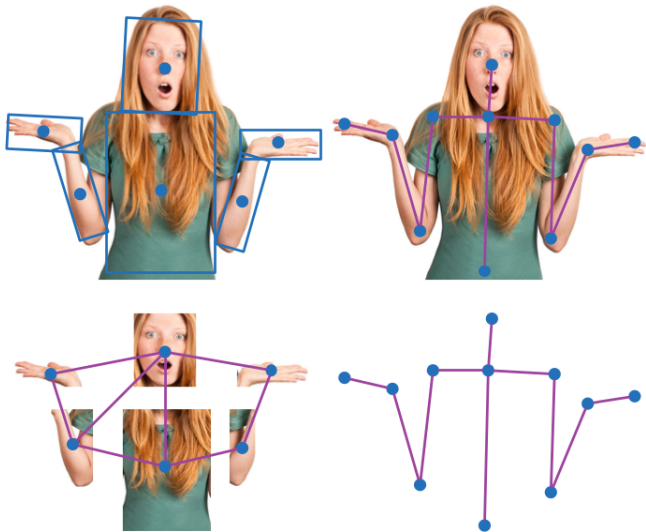
Application: gesture recognition



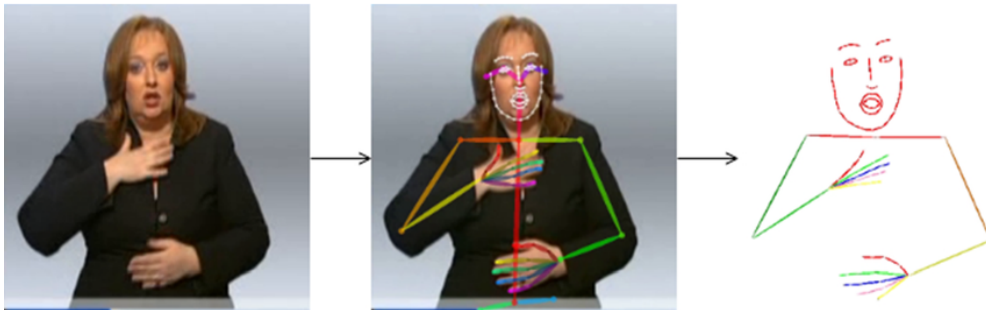
- May be used for gadget control

Application: emotion recognition

Robots need to identify emotions in order to interact with people properly



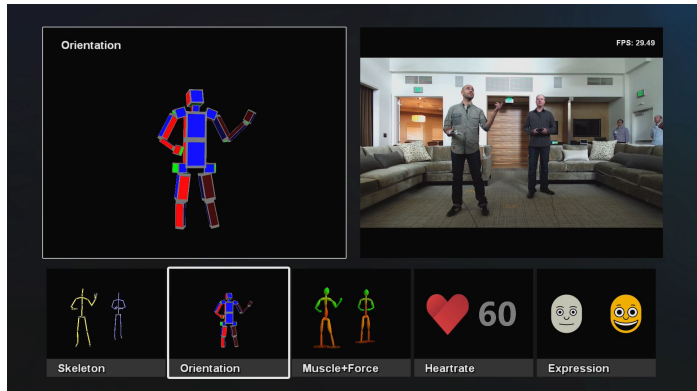
Application: sign language



- Sign language translators (<https://sign.mt/>)
- Sign language to voice

Application: games

PlayStation camera and Xbox Kinect allow to play Just Dance and Sports Games.



Label format

- Keypoint: set of coords (x, y) or (x, y, z)
- Skeleton connection: set of pairs $(joint_i, joint_j)$

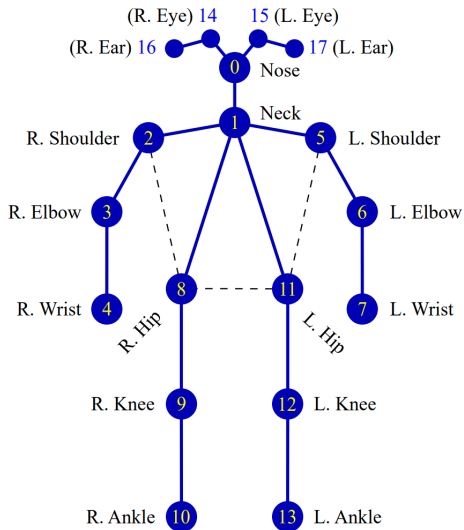


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MPII Human Pose

- 25K images containing over 40K people
- 410 human activities



COCO Pose



- Many iterations (2016-2020)
- Over **150k people** and **1.7 million labeled keypoints**

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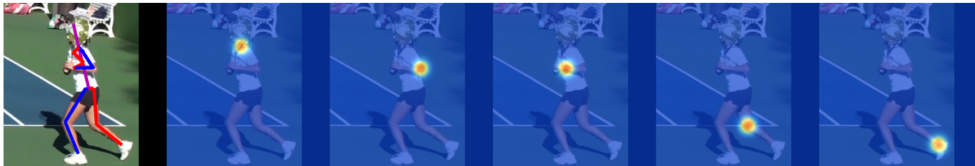
4. Metrics

DeepPose (2014)



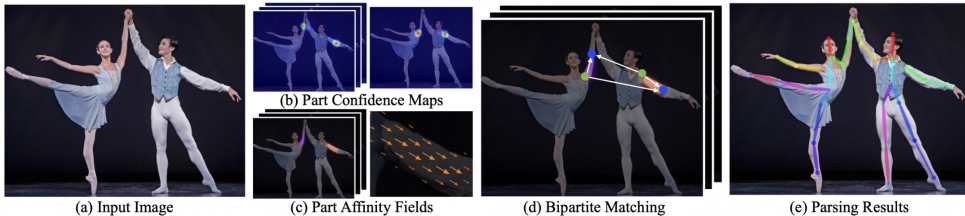
- Init stage: predict $2 * N$ coarse coords of keypoints
- Stage 1: take center crop of every point and predict refinement deltas
- Repeat refinement process several times

Heatmap approach



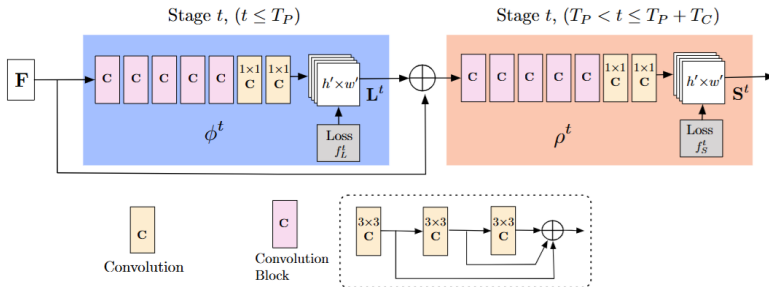
- Target heatmaps: gaussian centered in joint locations
- Loss: MSE between pred and gt intensities
- Point extraction: point with maximum intensity (for example)
- Architectures: Stacked Hourglass Networks, HRNet

OpenPose idea



- Part Confidence Maps: the same as in heatmap approach
- Part Affinity Fields: 2D vector field for each pair of keypoint
- Point extraction: point with maximum intensity (for example)

OpenPose architecture



- $\mathbf{S}^t \in \mathbb{R}^{J \times H \times W \times 1}$, J - number of keypoints
- $\mathbf{L}^t \in \mathbb{R}^{C \times H \times W \times 2}$, C - number of limbs (connections)
- $\mathbf{L}^1 = \phi^1(\mathbf{F})$, $\mathbf{F}$ - features from VGG
- $\mathbf{L}^t = \phi^t(\mathbf{F}, \mathbf{L}^{t-1})$, $\forall 2 \leq t \leq T_P$
- $\mathbf{S}^{T_P} = \rho^t(\mathbf{F}, \mathbf{L}^{T_P})$, $\forall t = T_P$,
- $\mathbf{S}^t = \rho^t(\mathbf{F}, \mathbf{L}^{T_P}, \mathbf{S}^{t-1})$, $\forall T_P < t \leq T_P + T_C$,

OpenPose Losses

$$f_L^t = \sum_{c=1}^C \sum_{\mathbf{p}} \mathbf{W}(\mathbf{p}) \cdot \|\mathbf{L}_c^t(\mathbf{p}) - \mathbf{L}_c^*(\mathbf{p})\|_2^2$$

$$f_S^t = \sum_{j=1}^J \sum_{\mathbf{p}} \mathbf{W}(\mathbf{p}) \cdot \|\mathbf{S}_j^t(\mathbf{p}) - \mathbf{S}_j^*(\mathbf{p})\|_2^2,$$

$$f = \sum_{t=1}^{T_P} f_L^t + \sum_{t=T_P+1}^{T_P+T_C} f_S^t.$$

where

- * means groundtruth
- $\mathbf{W}$ is a binary mask with $\mathbf{W}(\mathbf{p}) = 0$ when the annotation is missing at the pixel $\mathbf{p}$.
- T_P - number of stages for PAF, T_C - number of stages for confidence maps
- $\mathbf{p}$ - pixels
- J - number of keypoints, C - number of connections

OpenPose connection strength

If we consider a single pair of parts j_1 and j_2 (e.g., neck and right hip)

$$E = \int_{u=0}^{u=1} \mathbf{L}_c(\mathbf{p}(u)) \cdot \frac{\mathbf{d}_{j_2} - \mathbf{d}_{j_1}}{\|\mathbf{d}_{j_2} - \mathbf{d}_{j_1}\|_2} du,$$

where $\mathbf{p}(u)$ interpolates the position of the two body parts d_{j_1} and d_{j_2} ,

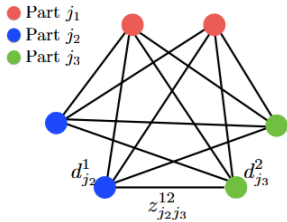
$$\mathbf{p}(u) = (1 - u)\mathbf{d}_{j_1} + u\mathbf{d}_{j_2}.$$

$$E \approx \frac{1}{L} \sum_{k=1}^L \mathbf{L}_c(x_k, y_k) \cdot \frac{\mathbf{d}_{j_2} - \mathbf{d}_{j_1}}{\|\mathbf{d}_{j_2} - \mathbf{d}_{j_1}\|_2}$$

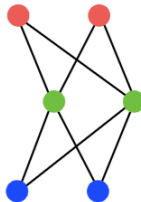
OpenPose matching



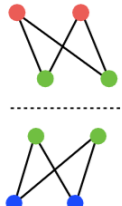
(a)



(b)



(c)



(d)

- Set of independent bipartite matching subproblems and get separate matchings for each limb
- Use Hungarian algorithm to match one limb
- Start with one matched connection (limb) and grow the skeleton by adding other connected limbs until no more connections are found

Mask R-CNN

- Predicts K masks, one for each of K keypoint types (e.g., left shoulder, right elbow).

Mask R-CNN Architecture

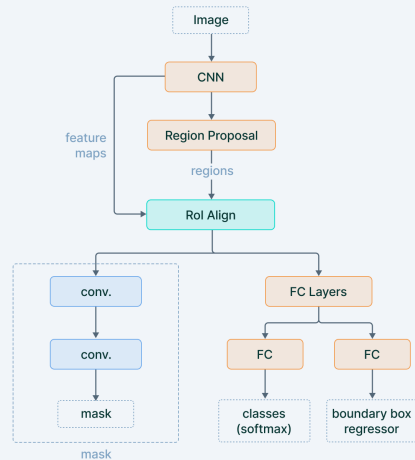


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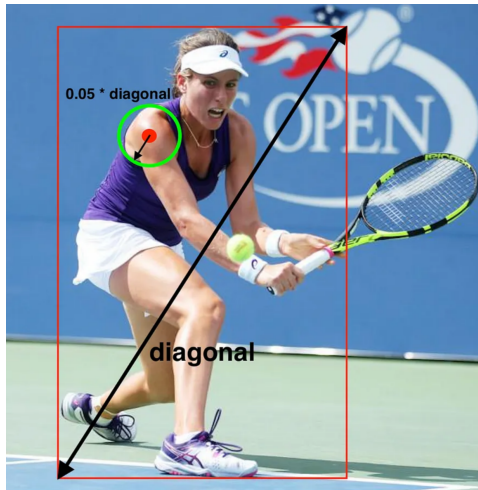
3. Architectures

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Percentage of Detected Joints (PDJ)

$$d_j = \sqrt{(x_j - x_j^*)^2 + (y_j - y_j^*)^2}$$

$$\text{PDJ} = \frac{1}{N} \sum_{j=1}^N \mathbb{I}(d_j \leq 0.05 \cdot \text{diagonal})$$

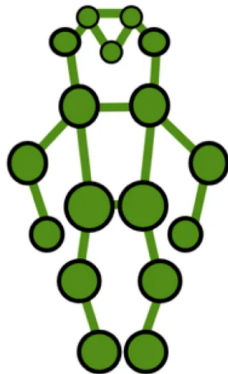


Object Keypoint Similarity (OKS)

$$\text{OKS} = \frac{\sum_{i=1}^K \exp\left(-\frac{d_i^2}{2s^2\kappa_i^2}\right) \delta(v_i > 0)}{\sum_{i=1}^K \delta(v_i > 0)}$$

$\delta(v_i > 0)$ - flag (was annotated)

Keypoint	k_i
hips	0.107
ankles	0.089
knees	0.087
shoulders	0.079
elbows	0.072
wrists	0.062
ears	0.035
nose	0.026
eyes	0.025



Percentage of Correct Parts (PCP)

$$d_{k_1} = \|\mathbf{p}_{k_1} - \mathbf{p}_{k_1}^*\|_2$$

$$d_{k_2} = \|\mathbf{p}_{k_2} - \mathbf{p}_{k_2}^*\|_2$$

$$l^* = \|\mathbf{p}_{k_2}^* - \mathbf{p}_{k_1}^*\|_2$$

$$\text{PCP} = \frac{\sum_{\text{Limb Set}} \mathbb{I}(d_{k_1} \leq 0.5 \cdot l^* \wedge d_{k_2} \leq 0.5 \cdot l^*)}{\text{Total Number of Limbs}}$$





Thank you for your time

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